

QUANTUM COMPUTING · BEGINNER NOTES

From Amplitudes to the H Gate

Understanding how quantum amplitudes become probabilities and how H creates superposition

11 Amplitudes → Probabilities

We have already seen that a qubit can be written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

The numbers α and β are called **amplitudes**.

But amplitudes are not the same thing as probabilities. To find the probability of a measurement result, we take the **magnitude squared** of its amplitude.

1. The most important rule

$$P(0) = |\alpha|^2$$

$$P(1) = |\beta|^2$$

This is known as the **Born rule**.

Remember:

Amplitude → square its magnitude → probability

Never confuse the amplitude itself with the probability.

2. Why do we square the amplitude?

Quantum mechanics uses amplitudes to describe quantum states. These amplitudes can even be **negative or complex**.

A probability, however, must always be a real number between **0 and 1**.

Squaring the magnitude converts the amplitude into a valid probability.

$$0 \leq P \leq 1$$

3. Simple example: amplitude = 1

Consider the state:

$$|\psi\rangle = 1|0\rangle$$

The amplitude of $|0\rangle$ is 1.

$$P(0) = |1|^2 = 1 = 100\%$$

Therefore, measuring this state gives 0 with certainty.

4. Simple example: amplitude = 0

Consider:

$$|\psi\rangle = 0|0\rangle + 1|1\rangle$$

The amplitude of $|0\rangle$ is 0:

$$P(0) = |0|^2 = 0$$

So there is no chance of measuring 0. The result will definitely be 1.

5. The 50–50 example

Consider the state:

$$|\psi\rangle = 1/\sqrt{2} |0\rangle + 1/\sqrt{2} |1\rangle$$

The amplitudes are:

$$\alpha = 1/\sqrt{2} \quad \beta = 1/\sqrt{2}$$

Now square them:

$$P(0) = |1/\sqrt{2}|^2 = 1/2 = 50\%$$

$$P(1) = |1/\sqrt{2}|^2 = 1/2 = 50\%$$

6. A more interesting example

Suppose:

$$|\psi\rangle = \sqrt{3}/2 |0\rangle + 1/2 |1\rangle$$

Calculate the probability of each result.

Probability of 0

$$P(0) = |\sqrt{3}/2|^2$$

$$= 3/4$$

$$= 75\%$$

Probability of 1

$$P(1) = |1/2|^2$$

$$= 1/4$$

$$= 25\%$$

7. Probabilities must add to 1

Since the qubit must produce one of the possible measurement outcomes, all probabilities must add up to 1.

$$|\alpha|^2 + |\beta|^2 = 1$$

For our previous example:

$$75\% + 25\% = 100\%$$

8. What if an amplitude is negative?

Consider:

$$|\psi\rangle = 1/\sqrt{2} |0\rangle - 1/\sqrt{2} |1\rangle$$

The amplitude of $|1\rangle$ is negative:

$$\beta = -1/\sqrt{2}$$

Its measurement probability is still:

$$P(1) = |-1/\sqrt{2}|^2 = 1/2 = 50\%$$

Important:

A negative amplitude does not mean a negative probability. Probabilities can never be negative.

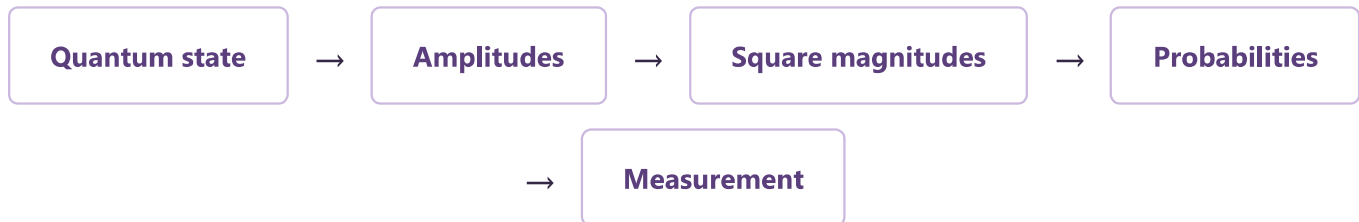
The sign of an amplitude becomes important when quantum states interact and interfere with one another.

9. Amplitude vs probability

Concept	Amplitude	Probability
Symbol	α or β	$P(0)$ or $P(1)$
Can be negative?	Yes	No
Can be complex?	Yes	No

Range	Not restricted to 0–1	0 to 1
Connection	Describes the quantum state	Determines measurement chances
Formula	α, β	$ \alpha ^2, \beta ^2$

10. The complete path



In one line:

Amplitudes describe the quantum state; their squared magnitudes determine the probabilities we observe when measuring.

Key idea

The probability of measuring a particular basis state is the squared magnitude of its amplitude. For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, the probabilities are $|\alpha|^2$ and $|\beta|^2$, and they must add up to 1.

12 Meet the H Gate

The **Hadamard gate**, usually written as **H**, is one of the most important gates in quantum computing.

One of its most useful jobs is to create **superposition** from a computational-basis state.

1. What is a quantum gate?

A quantum gate is an operation that changes the state of one or more qubits.

You can think of a gate as an instruction applied to a qubit.



2. The H gate

The Hadamard gate is represented by a square containing the letter H:



Its main beginner-level role is:

Definite state $\rightarrow$ Superposition

3. H acting on $|0\rangle$

The most important H-gate equation is:

$$H|0\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

This resulting state is called $|+\rangle$.

$$|+\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

4. What does that mean?

Before applying H, the qubit is definitely in $|0\rangle$.

After applying H, the state contains equal amplitudes for $|0\rangle$ and $|1\rangle$.



5. Now convert amplitudes into probabilities

The amplitude of $|0\rangle$ is:

$$\alpha = 1/\sqrt{2}$$

So:

$$P(0) = |1/\sqrt{2}|^2 = 1/2 = 50\%$$

Similarly:

$$P(1) = |1/\sqrt{2}|^2 = 1/2 = 50\%$$

Therefore:

Applying H to $|0\rangle$ creates a state where a computational-basis measurement gives 0 or 1 with equal probability.

6. The H gate as a circuit



Start in $|0\rangle \rightarrow$ apply H $\rightarrow$ measure

Repeating this circuit many times should produce approximately equal numbers of 0 and 1.

7. H acting on $|1\rangle$

The H gate can also act on $|1\rangle$:

$$H|1\rangle = (|0\rangle - |1\rangle) / \sqrt{2}$$

This state is called $|-\rangle$.

$$|-\rangle = (|0\rangle - |1\rangle) / \sqrt{2}$$

8. Notice the minus sign

Compare the two states:

$$H|0\rangle$$

$$(|0\rangle + |1\rangle) / \sqrt{2}$$

H|1⟩

$$(|0\rangle - |1\rangle) / \sqrt{2}$$

The sign changes the **amplitude**, but not the individual computational-basis measurement probabilities.

$$|+\rangle \rightarrow 50\% 0, 50\% 1$$

$$|-\rangle \rightarrow 50\% 0, 50\% 1$$

The relative sign becomes important when the state later interferes with other quantum states.

9. The H gate matrix

Quantum gates can be represented using matrices. The Hadamard gate is:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

This matrix acts on the qubit's state vector.

10. Matrix calculation for H|0⟩

Represent $|0\rangle$ as:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Apply H:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Multiplying gives:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Which is exactly:

$$H|0\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

11. H is reversible

One beautiful property of the Hadamard gate is that applying it twice brings the original state back.

$$H^2 = I$$

Here **I** is the identity operation, which means "do nothing."



12. H followed by H in a circuit



H twice returns the qubit to its original state

$$H(H|0\rangle) = |0\rangle$$

13. H vs X

Gate	Action on $ 0\rangle$	Result
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X	$X 0\rangle$	$ 1\rangle$
H	$H 0\rangle$	$(0\rangle + 1\rangle)/\sqrt{2}$

Think of the difference this way:

X: flips the state.

H: creates an equal superposition from a basis state.

14. Why is the H gate important?

The H gate appears in many quantum algorithms because it provides an easy way to move between computational-basis states and superposition states.

It is especially useful for:

- Creating superposition
- Preparing qubits for quantum algorithms
- Producing interference when combined with other gates
- Converting between different quantum-state bases

15. The complete picture



$$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

↓

$$P(0) = 50\% \quad P(1) = 50\%$$

Key idea

The Hadamard gate transforms a basis state into a superposition. In particular, $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, which produces 0 and 1 with equal probability when measured in the computational basis. Applying H twice

returns the original state because $H^2 = I$.

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Quick Recap

Concept	Remember
Amplitude	Describes a component of the quantum state
Probability	Squared magnitude of the corresponding amplitude
Born rule	$P = \text{amplitude} ^2$
Hadamard gate	Creates superposition from a basis state
$H 0\rangle$	$(0\rangle + 1\rangle)/\sqrt{2} = +\rangle$
$H 1\rangle$	$(0\rangle - 1\rangle)/\sqrt{2} = -\rangle$
H^2	I — applying H twice returns the original state